

Aarti Jadhav

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EDUCATION

Master of Science in Data Science

University at Buffalo, The State University of New York

GPA: 3.630/4.0

August 2024 – December 2025

Bachelor of Technology in Computer Science

D. Y. Patil College of Engineering and Technology, Kolhapur, India

GPA: 8.50/10.0

July 2020 – April 2024

SKILLS

Programming Languages: Python, R, Java, SQL, ABAP, MATLAB, JavaScript, C++, Ruby.

Machine Learning Frameworks: Scikit-learn, TensorFlow, OpenCV, PyTorch, Keras, XGBoost, LightGBM, Transfer Learning.

Optimization Tools: PuLP, Gurobi, CPLEX, Linear Programming, Jira, Git, Eclipse IDE for ABAP, SAP GUI, Hyperparameter Tuning, Grid Search.

Database Management Systems: RDBMS, Azure SQL, MySQL, Databricks, MongoDB, Oracle DB, BigSQL, NoSQL, Hadoop, Spark, Hive.

Visualization Tools: Power BI, Tableau, Matplotlib, Seaborn, Google Data Studio, Excel, Microsoft Power Map.

Cloud & Software: SAS, Apex, Microsoft Azure, Excel, AWS, Google Cloud, RStudio, Google Colab, Jupyter.

Interpersonal Skills: Collaboration, Communication, Leadership, Attention to Detail, Critical Thinking, Problem Solving, Data Storytelling, Adaptability.

Other Skills: Statistical Data Analysis, Data Wrangling, Query Writing, Reporting & Analysis, Data Governance.

Languages: English, German, Hindi, Marathi.

PROFESSIONAL EXPERIENCE

Data Scientist Intern

August 2023 – April 2024

Kolhapur Cancer Center, Kolhapur, India

- Research Topic:** Investigating the Application of Artificial Intelligence and Machine Learning (AIML) Techniques to Optimize Electrochemotherapy for Cancer Treatment.
- Expedited electrochemotherapy treatment efficacy by 20% through implementation of Random Forest and Gradient Boosting models, optimizing electrical parameters and delivering enhanced tumor regression rates for 500+ patients.
- Implemented a machine learning pipeline for parameter optimization, integrating clinician feedback to enhance tumor regression rates by 15%. Collaborated with oncologists and clinical teams to ensure practical application of AI-driven insights in treatment workflows.

Product Development & Tech Support Intern

December 2023 – April 2024

RN Software & Consultants, Kolhapur, India

- Engineered a predictive machine learning model using XGBoost to forecast hotel room demand, increasing marketing efficiency by 30%. Analyzed three years of historical data (50,000+ records) for seasonal trends, enabling adaptive pricing strategies and boosting seasonal revenue by 12%.
- Discovered and deployed interactive dashboards using Tableau for real-time monitoring and strategy optimization. Integrated advanced analytics tools into existing hotel management software to streamline operations and improve decision-making.

PROJECTS

Crime Trends and Incarceration Analysis

- Analyzed 15 years of crime data (500,000+ records) using Python to uncover trends in violent crimes and incarceration rates. Conceived visualizations using Python and Streamlit, including dynamic dashboards with Plotly for trend analysis.
- Designed a normalized SQL database schema (6 tables) to structure data for querying and deeper insights. Defined interactive dashboards in Streamlit with Plotly for real-time trend visualization and reporting.
- Key Insights: Violent crimes dominate in the 18-30 age group, with strong correlations observed between prison capacity and inmate population.

Optimal Solar Energy Usage Scheduling

- Streamlined a Mixed Integer Programming (MIP) model to optimize the scheduling of household appliances based on solar energy generation peaks. The goal was to maximize energy efficiency by aligning appliance usage with available solar energy while minimizing reliance on the grid.
- Gathered historical solar generation data and household energy consumption patterns. Designed a MIP model using PuLP and solved using Gurobi, which incorporated energy consumption profiles, solar generation forecasts, and appliance priorities.
- Applied advanced scheduling algorithms to allocate appliances during optimal solar energy generation periods. Achieved a 25% increase in energy efficiency by optimizing appliance usage, reducing electricity bills and minimizing grid dependency.

Datamind- Data Analysis Tool

- DataMind is an advanced interactive data analysis tool that leverages cutting-edge AI technologies to provide intuitive data exploration and visualization. Built with Streamlit and LangChain, this application implements multi-agent systems with chain-of-thought prompting, enabling dynamic decision-making and task routing.
- It features stateful memory management with chat history injection, interactive data visualization using Plotly, and a conversational AI system designed for seamless user interaction.

Sales-Forecasting-TSA

- Designing a Time Series forecasting model to assess and predict the pattern of Sales of an online E-commerce website. Involves in-depth data analysis along with exploration and implementation of various important Time Series Analysis (TSA) techniques.

ACHIEVEMENTS

- Finalist, IIT Bombay eSummit Hackathon, Piramal Data Science Track.
- Published paper on "Twitter Sentiment Analysis" in IJISRT.
- Former Technical Co-Lead, Google Developer Student Clubs (GDSC).
- Awarded by The Greater Bombay Science Teachers' Association for excellence in Dr. Homi Bhabha Balvaidnyanik Competition.